

Software Requirements Specification (SRS)

Project Title: Color Blindness Detection using Ishihara Plates

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a detailed description of the functional and non-functional requirements of the VisionCheck AI – Color Blindness Detection System using Ishihara Plates. The primary purpose of this system is to assist in the early identification of color vision deficiencies by leveraging Ishihara testing methods, machine learning algorithms, and intelligent web-based interfaces.

Color vision deficiency affects a significant portion of the population and often remains undetected due to the lack of accessible screening tools. Traditional manual testing methods require supervision and may not provide scalable or data-driven analysis. This system addresses these challenges by offering an interactive and intelligent platform that enables users to perform tests remotely while ensuring efficient data processing, result visualization, and analytical reporting.

This document serves as a formal agreement between stakeholders and developers regarding system expectations, design constraints, and operational capabilities. It is intended for:

- Developers responsible for designing and implementing the frontend, backend, and machine learning components
- Testers who validate system functionality, performance, and prediction accuracy
- Project evaluators and academic reviewers assessing system design and completeness
- System administrators responsible for deployment, monitoring, and maintenance of cloud infrastructure

The SRS defines system functionality, interfaces, constraints, performance expectations, and quality requirements. It acts as a baseline reference for system development, testing, deployment, and future enhancements, ensuring that the VisionCheck AI system is reliable, scalable, and maintainable throughout its lifecycle.

1.2 Scope

The VisionCheck AI system is designed to operate as an intelligent web-based screening platform for identifying color vision deficiencies using Ishihara plates and machine learning techniques. Its primary objective is to provide an accessible and automated environment where users can perform color vision tests and receive accurate classification results without the need for manual supervision. By integrating interactive interfaces with predictive analytics, the system aims to improve the efficiency, scalability, and reliability of color blindness screening.

The system focuses on analyzing user responses to Ishihara test plates and identifying patterns that correspond to different types of color vision conditions, including Normal Vision, Protanopia, Deutanopia, and Tritanopia. Through the application of machine learning algorithms and behavioral response analysis, the platform aims to uncover subtle variations in visual perception that may not be easily detected through traditional rule-based or manual testing approaches.

VisionCheck AI provides automated test execution, real-time prediction of vision type, and data-driven insights through analytics and visualization tools. It maintains historical records of test responses, enabling trend analysis and performance monitoring. The system also incorporates MLOps practices to support continuous model improvement, allowing administrators to retrain models using updated datasets and evaluate prediction performance over time.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
ML	Machine Learning
UI	User Interface

API	Application Programming Interface
MLOps	Machine Learning Operations
Ishihara Plate	Color vision test image

1.4 Overview

This SRS document is organized as follows:

- Section 2 provides an overall description of the system, including its architecture, operating environment, user roles, assumptions, and design constraints.
- Section 3 defines external interface requirements, covering user interfaces, hardware interfaces, software integrations, and communication mechanisms used by the system.
- Section 4 describes system features and functional requirements such as Ishihara test management, vision classification, response handling, analytics, and model training capabilities.
- Section 5 outlines non-functional requirements including performance expectations, security measures, scalability considerations, usability standards, and maintainability guidelines.
- Section 6 specifies safety requirements to ensure responsible usage of prediction results and protection against incorrect or misleading outputs.
- Section 7 includes additional system requirements such as explainable AI features, logging mechanisms, and future extensibility considerations.
- Appendices provide glossary definitions, supporting diagrams, and additional reference materials relevant to system design and analysis.

1.5 References

The system design is based on research in:

- IEEE 830-1998 SRS Standard
- Research papers on Ishihara Color Blindness Testing
- Machine Learning Classification Documentation

2. Overall Description

2.1 Product Perspective

The VisionCheck AI system is a data-driven intelligent web application designed to operate as an automated color vision screening platform. It does not replace professional medical diagnosis but functions as an intelligent assessment and decision-support system that assists users in identifying potential color vision deficiencies through Ishihara plate testing and machine learning analysis.

The system integrates with external platforms and services such as:

- Cloud storage services for storing Ishihara images and trained machine learning models
- Machine learning frameworks used for model training and inference
- Web-based analytics and visualization libraries
- Cloud deployment infrastructure for scalable system operation

The system architecture consists of the following subsystems:

1. Frontend Layer

Provides an interactive dashboard where users can:

- View Ishihara test plates
- Submit responses during the test

- Receive predicted vision type results
- Receive predicted vision type results
- Access visual analytics and reports

2. Application/Backend Layer

Responsible for:

- API communication between frontend and backend services
- Processing and validating user responses
- Managing authentication, user sessions, and access control
- Handling data storage and retrieval operations
- Generating analytics data and reports

3. Vision Classification Engine

This module:

- Applies trained machine learning models for classification
- Analyzes response patterns from Ishihara tests
- Predicts vision categories such as Normal Vision, Protanopia, Deuteranopia, and Tritanopia
- Generates prediction scores and classification outputs

4. Analytics and Visualization Module

This module:

- Generates statistical insights from collected test data
- Displays charts and graphs representing vision type distribution

- Supports trend analysis and performance monitoring
- Assists administrators in understanding system usage patterns

5.Database Layer

Stores:

- User test responses and prediction history
- Model versions and configuration data
- System logs and analytics data
- Ishihara plate images and trained model artifacts

The VisionCheck AI system is designed as a scalable and modular platform capable of supporting continuous model improvement, cloud-based deployment, and future integration of advanced machine learning techniques and analytical features.

2.2 Product Functions

The system provides the following major functional capabilities:

1. Ishihara Test Execution

The system presents Ishihara plates to users in a structured sequence through an interactive web interface. It manages the test flow by displaying images, capturing user responses, and ensuring that inputs are validated before processing.

2. Response Collection and Management

The system collects user inputs corresponding to each Ishihara plate and organizes them into structured datasets. It verifies response integrity, handles incomplete or incorrect entries gracefully, and prepares the collected data for further analytical processing and prediction.

3. Feature Processing and Data Preparation

The system transforms raw user responses into meaningful analytical features required for machine learning classification.

4. Vision Classification and Prediction

The system applies trained machine learning models to evaluate user responses and predict vision type. It generates classification outputs such as Normal Vision, Protanopia, Deuteranopia, or Tritanopia along with prediction scores, enabling accurate and data-driven assessment of color vision conditions.

5. Result Management and History Tracking

The system stores prediction results and user test history in a database for future reference. Users can view past test results, while administrators can analyze historical data to understand system performance and user trends.

6. Analytics and Visualization

The system produces analytical summaries and visual reports, including charts and statistical insights related to vision type distribution, user participation, and prediction trends.

7. Model Lifecycle Management

The system supports continuous improvement of machine learning models through retraining and version management. It maintains model history, evaluates performance metrics, and enables integration with MLOps workflows to ensure consistent prediction accuracy and adaptability to new datasets.

2.3 User Classes and Characteristics

User Class	Role in System	System Interaction	Technical Skill
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General User	Performs color vision test using Ishihara plates	Views test interface, submits responses, views prediction results and history	Low
Administrator	Monitors system analytics and manages test content	Accesses analytics dashboard, reviews reports, manages Ishihara plates and user data	Moderate
System Administrator	Maintains system	Configures system, manages database and models	High

2.4 Operating Environment

The system operates under the following environment:

- **Hardware:** Cloud-based servers for application hosting and data storage
- **Operating Systems:** Linux (primary), Windows (secondary)
- **Frontend Technologies:** React.js
- **Backend Technologies:** Spring Boot
- **ML Frameworks:** Scikit-learn
- **Database:** SQL
- **Cloud:** AWS S3, Ec2

Containers: Docker

Orchestration: Kubernetes

2.5 Design and Implementation Constraints

The following constraints impact system development:

- **Computational Constraint:** Machine learning inference and analytics processing require adequate computing resources for efficient performance
- **Data Constraint:** Training datasets for color vision classification may be limited or imbalanced, affecting model accuracy
- **Privacy Constraint:** User test responses and personal data must be handled securely to maintain confidentiality
- **Cloud Dependency Constraint:** System functionality depends on cloud services such as AWS S3 and database availability
- **Scalability Constraint:** The system must support increasing numbers of users and growing historical test data
- **Security Constraint:** Secure authentication, encrypted communication, and access control mechanisms must be implemented

2.6 Assumptions and Dependencies

Assumptions

- Users have basic familiarity with web interfaces and can follow Ishihara test instructions
- Ishihara plate images used in the system are accurate and properly configured
- Machine learning models are pre-trained and available for prediction

Dependencies

- Machine learning libraries such as Scikit-learn, TensorFlow, or PyTorch

- MongoDB database infrastructure for storing responses and results
- Cloud services such as AWS S3 for storage and deployment

3. External Interface Requirements

3.1 User Interfaces

The VisionCheck AI system provides a web-based graphical user interface (GUI) that allows users to perform color vision tests, view prediction results, and manage analytical data through an interactive platform. The interface is accessible through modern web browsers and does not require installation of additional software, ensuring ease of access across multiple devices.

The primary interface is an Ishihara test dashboard where users can view test plates sequentially and submit their responses through a structured input form. The interface ensures smooth navigation between test stages and provides clear instructions to guide users throughout the screening process. Once the test is completed, the system displays prediction results along with visual indicators representing the classified vision type.

A results and history interface enables users to review past test records, prediction outcomes, and timestamps. Information is presented in an organized format with filtering and search capabilities, allowing users to quickly access specific test sessions. Visual elements such as charts and graphs may be included to help users understand their performance trends over time.

The system also provides an analytics dashboard for administrative users. This interface displays aggregated statistics such as vision type distribution, user participation trends, and system performance metrics. Interactive charts allow administrators to analyze data efficiently and generate summary reports that can be viewed on-screen or exported for further analysis.

Administrative users access a management interface where they can configure system settings, manage Ishihara plate content, monitor model performance, and control access permissions. All user interface components follow usability principles such as consistent layout, clear labeling, responsive design, and accessibility to ensure a smooth and intuitive user experience.

3.2 Hardware Interfaces

The VisionCheck AI system requires cloud-based server infrastructure capable of supporting web application hosting, database operations, and machine learning inference. Backend servers must provide sufficient CPU resources to process user requests, manage data storage, and perform real-time prediction tasks efficiently. Adequate memory capacity is required to handle concurrent users and maintain smooth system performance during test execution and analytics processing.

Optional GPU hardware may be used to accelerate machine learning model training and performance evaluation processes, especially when handling larger datasets or advanced models. Storage infrastructure must support scalable data management, including user responses, prediction history, analytics data, and trained model artifacts stored in cloud services.

3.3 Software Interfaces

The VisionCheck AI system interfaces with frontend and backend components through secure RESTful APIs. The React-based frontend communicates with the Flask backend using structured JSON data formats to transmit user responses, retrieve prediction results, and access analytics information. These service interfaces ensure reliable and efficient data exchange between client and server components.

Internally, the system integrates with machine learning frameworks for model training and inference. The backend interacts with machine learning libraries to process user response data, generate classification outputs, and evaluate prediction performance. Model artifacts and related datasets may be stored and retrieved from cloud storage services as required.

The system also interfaces with a MongoDB database management system to store user responses, prediction results, analytics data, system logs, and configuration settings. Additionally, logging and monitoring tools may interface with the backend to capture operational metrics and maintain system performance tracking.

3.4 Communication Interfaces

The VisionCheck AI system uses secure communication channels for all data exchanges between system components. Data transmission between the React-based frontend and the Flask backend is encrypted using HTTPS protocols to prevent unauthorized access and ensure data confidentiality. Communication with cloud services such as storage platforms and machine learning resources also occurs through secure encrypted connections.

The system supports real-time data transfer to process user responses and generate prediction results with minimal delay. Authentication and authorization mechanisms regulate access to communication endpoints. Only authenticated users and authorized services are permitted to interact with system APIs, ensuring protection against unauthorized access, data manipulation, or misuse of system resources.

4. System Features and Functional Requirements

4.1 Transaction Monitoring

Description

This feature enables the VisionCheck AI system to present Ishihara test plates to users and collect their responses through an interactive web interface. It ensures that all user inputs are validated, structured, stored, and prepared for machine learning prediction. The test management module serves as the entry point of the system and must operate reliably to provide a smooth and consistent testing experience.

Stimulus / Response

- **Stimulus:** A user starts a color vision test and submits responses to displayed Ishihara plates.
- **Response:** The system validates the responses, stores the data, and forwards it to the prediction module for analysis.

Functional Requirements

- **REQ-1:** The system shall display Ishihara plates sequentially to the user.
- **REQ-2:** The system shall accept and validate user responses for each plate.
- **REQ-3:** The system shall handle missing or invalid inputs gracefully.
- **REQ-4:** The system shall preprocess collected responses prediction processing.
- **REQ-5:** The system shall store user responses in the database.
- **REQ-6:** The system shall allow retrieval of previous test records and history.

4.2 Fraud Detection Engine

Description

The vision classification engine analyzes user responses collected from Ishihara test plates and predicts the type of color vision condition using machine learning models. It applies data preprocessing, pattern recognition, and classification techniques to identify whether the user has Normal Vision, Protanopia, Deuteranopia, or Tritanopia..

Stimulus / Response

- **Stimulus:**Preprocessed user responses are forwarded to the classification engine for evaluation.
- **Response:** The system generates a prediction result and assigns a corresponding vision category along with a confidence score.

Functional Requirements

- **REQ-7:** The system shall extract relevant features from user responses for prediction processing.
- **REQ-8:** The system shall apply trained machine learning models to classify vision types.

- **REQ-9:** The system shall generate a prediction score representing classification confidence.
- **REQ-10:** The system shall classify transactions into risk categories.
- **REQ-11:** The system shall log prediction results for monitoring and auditing purposes.
- **REQ-12:** The system shall support model evaluation to handle diverse response patterns.

4.3 Graph Analysis Module

Description

This feature analyzes collected test data and generates visual insights to help administrators understand system usage patterns and vision type distribution.

Stimulus / Response

- **Stimulus:** New prediction results or user test data are stored in the system..
- **Response:** The analytics module updates charts, statistics, and visual reports accordingly.

Functional Requirements

- **REQ-13:** The system shall generate visual charts representing vision type distribution.
- **REQ-14:** The system shall update analytics dynamically when new test results are added.

- **REQ-15:** The system shall compute statistical summaries from stored prediction data.
- **REQ-16:** The system shall allow administrators to view interactive graphs and analytics dashboards.
- **REQ-17:** The system shall support exporting analytical reports in standard formats.

4.4 Alert and Notification System

Description

This feature notifies users when a vision classification result is generated after completing the Ishihara test. It provides clear feedback regarding the predicted vision type and allows users to review detailed results.

Stimulus / Response

- **Stimulus:** A user completes the Ishihara test and prediction processing is finished.
- **Response:** The system generates a result notification, displays the vision classification, and stores the result in history records.

Functional Requirements

- **REQ-18:** The system shall generate result notifications after prediction processing.
- **REQ-19:** The system shall display the classified vision type to the user.
- **REQ-20:** The system shall store result history in the database.
- **REQ-21:** The system shall allow users to view detailed result information.
- **REQ-22:** The system shall support acknowledgment or review of results by the user.

4.5 Reporting and Compliance Module

Description

The reporting module provides analytical insights based on vision classification results and historical test data.

Stimulus / Response

- **Stimulus:** A user or administrator requests a report or analytical summary.
- **Response:** The system generates, displays, or exports the requested report based on stored test data and analytics.

Functional Requirements

- **REQ-23:** The system shall generate summaries of vision classification results.
- **REQ-24:** The system shall provide trend analysis based on historical test data.
- **REQ-25:** The system shall support exporting reports in standard formats.
- **REQ-26:** The system shall maintain historical reporting records for analysis.

4.6 Model Training and Updating

Description

This feature ensures continuous improvement of fraud detection accuracy by retraining models and maintaining performance.

Stimulus / Response

- **Stimulus:** New labelled data or scheduled update.

Functional Requirements

- **REQ-27:** The system shall support retraining of ML models.
- **REQ-28:** The system shall maintain model version history.
- **REQ-29:** The system shall evaluate model accuracy after retraining.
- **REQ-30:** The system shall allow administrators to schedule retraining.

5. Non-Functional Requirements

This section describes the quality characteristics and operational constraints of the Cryptocurrency Fraud Transaction Detection System. These requirements ensure that the system performs reliably, securely, and efficiently in a real-world environment.

5.1 Performance Requirements

The system shall provide responsive and efficient performance during Ishihara test execution and prediction processing. Vision classification must occur quickly after user responses are submitted so that users receive results without noticeable delay. Real-time interaction between the frontend and backend components should ensure a smooth and uninterrupted testing experience.

The system shall be capable of handling multiple users simultaneously while maintaining stable response times for operations such as loading test plates, generating prediction results, viewing analytics dashboards, and retrieving historical test records. User interactions including navigation between test steps, viewing reports, and accessing analytics should remain within acceptable performance limits to ensure usability.

5.2 Security Requirements

Security is a critical requirement due to the sensitive nature of user response data and prediction results. The system shall implement strong authentication mechanisms to ensure that only authorized users can access system features. Role-based access control shall be enforced so that administrative privileges are restricted to authorized personnel.

All communication between frontend, backend, database, and cloud services shall be protected using secure transmission protocols such as HTTPS. Sensitive information including user responses, prediction results, and stored records shall be safeguarded against unauthorized access, modification, or disclosure.

5.3 Reliability Requirements

The system shall operate consistently with minimal downtime to ensure uninterrupted access to color vision testing services. It must be capable of handling unexpected failures without losing critical user data, prediction records, or model configurations. Recovery mechanisms shall allow the system to resume normal operations promptly after interruptions.

Regular data backups shall be maintained to support recovery in case of database corruption, hardware failure, or system errors. The system shall ensure integrity and consistency of stored responses, prediction outputs, and analytical data.

5.4 Scalability Requirements

The system shall be designed to accommodate increasing numbers of users and growing volumes of historical test data over time. It must support the addition of computing resources to handle higher processing loads for prediction and analytics tasks.

Database storage and cloud infrastructure shall be scalable to support long-term data growth. The system architecture shall allow distributed deployment or cloud-based scaling to maintain performance as system usage expands.

5.5 Usability Requirements

The system shall provide a user-friendly interface that enables users to complete Ishihara tests without requiring technical expertise. Prediction results, charts, and reports shall be presented clearly and logically to enhance understanding.

Navigation shall be intuitive, and the system shall guide users through each test step with clear instructions and visual indicators. The interface shall remain consistent and responsive across different devices and screen sizes to ensure accessibility and ease of use.

5.6 Maintainability Requirements

The VisionCheck AI system shall be developed using modular and scalable design principles to allow easy updates, maintenance, and future enhancements. Machine learning models shall be replaceable or upgradable without requiring significant changes to the overall system architecture.

System documentation, structured coding practices, and detailed logging mechanisms shall support efficient debugging, monitoring, and performance optimization. Proper version control and configuration management shall be maintained to ensure smooth integration of new features and updates.

5.7 Portability Requirements

The system shall be deployable across multiple operating systems and cloud environments to ensure flexibility and adaptability. It should not rely on a single hardware configuration or

deployment platform, allowing migration between development, testing, and production environments with minimal changes.

Containerization and cloud-based deployment strategies shall enable consistent performance across different infrastructure setups, ensuring that the application remains accessible and scalable regardless of hosting environment.

6. Safety Requirements

Although the VisionCheck AI system does not control physical hardware or perform medical diagnosis, it operates in a health-related assessment context where incorrect predictions or misleading outputs could cause confusion or improper interpretation by users. Therefore, the system must include safeguards to ensure safe and responsible operation.

The system shall prevent unsafe decisions caused by incorrect or misinterpreted prediction results. Vision classification outputs are intended to assist users as preliminary screening information and must not be presented as medical diagnoses. The system shall clearly communicate that results require human interpretation and professional consultation where necessary.

The system shall handle system failures in a controlled and predictable manner. In the event of software errors, network disruptions, or infrastructure failures, the system shall avoid generating incomplete or inconsistent prediction results. Instead, it shall notify administrators, log system issues, and temporarily suspend affected services until stability is restored.

Data integrity is a critical safety requirement. User responses, prediction results, and analytical records shall be protected against loss or corruption during unexpected shutdowns or crashes. Backup and recovery mechanisms shall ensure that the system can restore normal operations without compromising stored data consistency.

The system shall protect against accidental or unauthorized modification of model parameters and configuration settings. Administrative actions such as updating models, changing thresholds, or modifying system settings shall be restricted to authorized users to prevent unsafe configurations that could reduce prediction reliability.

The system shall maintain audit logs of critical events including model updates, configuration changes, prediction processing, and system errors. These logs support monitoring, accountability, and investigation of safety-related incidents.

Finally, the system shall ensure that prediction results do not automatically enforce decisions or restrict user actions. The platform functions as a supportive analytical tool, and any interpretation or decision-making must involve human oversight. The system shall also be designed to degrade gracefully, allowing unaffected modules to continue operating if a specific component encounters failure.

7. Other Requirements

This section includes additional requirements that do not fall directly under functional, performance, or safety categories but are essential for the effective operation, transparency, and long-term evolution of the VisionCheck AI system.

The system shall support explainable artificial intelligence mechanisms to provide insight into vision classification results. Users and administrators should be able to understand the reasoning behind prediction outputs, which improves transparency, usability, and trust in the system.

The system shall maintain comprehensive audit logs of user interactions, prediction outcomes, model updates, configuration changes, and system events. These logs support traceability, performance monitoring, and system analysis for continuous improvement.

The system shall follow responsible data usage practices and applicable privacy guidelines to ensure that user information is handled securely. Reporting and analytics features should present results clearly while maintaining data confidentiality.

The system architecture shall support future expansion to include improved machine learning models, additional analytics features, and integration with new datasets or visualization tools. The design must allow the addition of new modules without requiring major structural redesign.

The system shall provide data export capabilities that allow administrators to download reports and analytical summaries for external evaluation or research purposes. Exported data shall follow structured and standardized formats to ensure compatibility with other analytical tools.

The system shall maintain updated documentation and configuration records to support maintenance, debugging, and future enhancements. All system updates shall be traceable through version control and proper documentation practices

Appendix A – Glossary

Term	Definition
Ishihara Plate	A visual test image used to identify color vision deficiencies by displaying numbers within colored dot patterns.
Vision Classification	The predicted category of a user's color vision condition such as Normal Vision, Protanopia, Deutanopia, or Tritanopia.
Prediction Score	A numerical value representing the confidence level of a classification result..

Machine Learning (ML)	A technique where systems learn patterns from data to make predictions.
Dataset	A visual interface that displays charts, statistics, and system insights.
Analytics Dashboard	A wallet address represented as a point in a transaction graph.
User Response	Input provided by a user during the Ishihara test process.
Model Retraining	The process of updating machine learning models using new data to improve performance.
Cloud Storage	Remote storage service used to store images, datasets, and model files.

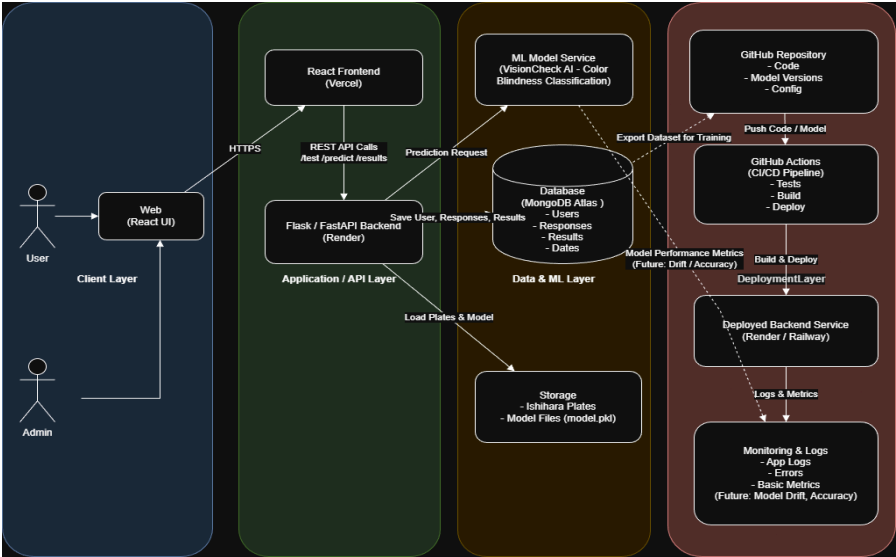
Appendix B – Analysis Models

This appendix contains diagrams that describe the structural design, data flow, and operational behavior of the VisionCheck AI system. These models support understanding of system architecture, processing logic, and interaction between different system components.

B.1 System Architecture Diagram

This diagram illustrates the high-level architecture of the VisionCheck AI platform. It presents the major subsystems including the React-based frontend interface, Flask backend services, machine learning classification engine, database layer, and cloud storage components.

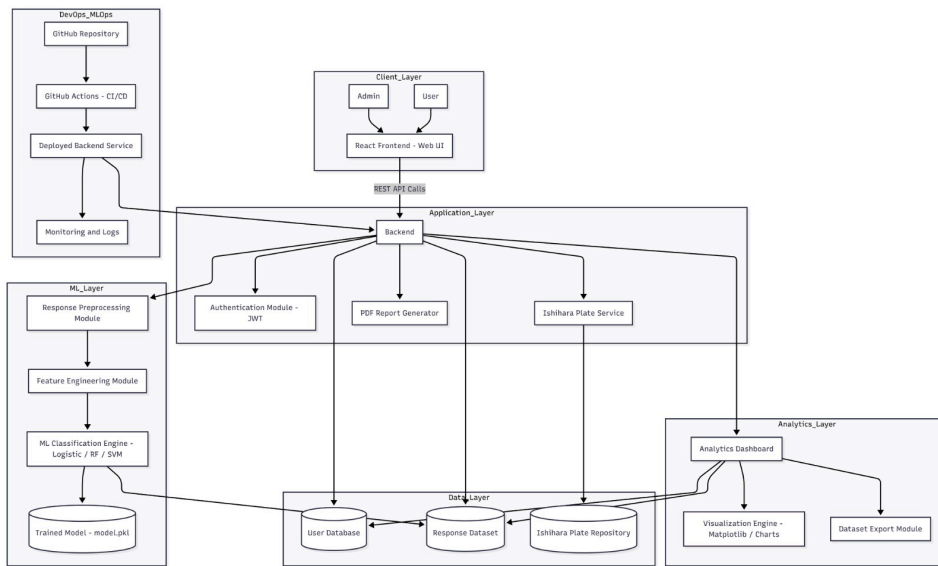
The architecture diagram shows how user responses flow from the frontend interface to the backend processing layer, where data validation and preprocessing occur before being forwarded to the vision classification engine. Prediction results are stored in the database and made available to analytics and reporting modules.



B.2 Component Diagram

This diagram represents the software components and services that make up the VisionCheck AI system. It illustrates the interaction between the React-based frontend modules, backend API services, authentication mechanisms, machine learning classification engine, analytics module, and supporting database and cloud storage layers.

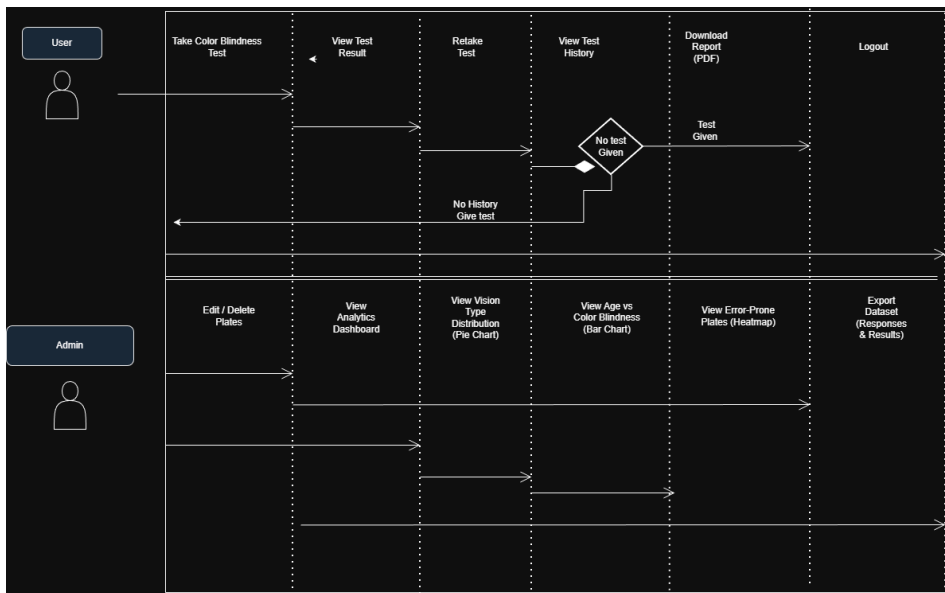
The component diagram includes frontend interface modules responsible for displaying Ishihara plates and collecting user responses, backend API services that handle data processing and communication, and the vision classification service that performs machine learning inference. Supporting components such as the MongoDB database, cloud storage services for model artifacts and images, and logging or monitoring tools are also represented.



B.3 Sequence Diagram

This diagram shows the step-by-step interaction between system components when a user performs an Ishihara color vision test and submits responses. It represents the sequence of operations starting from the frontend request, backend validation and processing, machine learning prediction, and returning the classified vision result to the user interface.

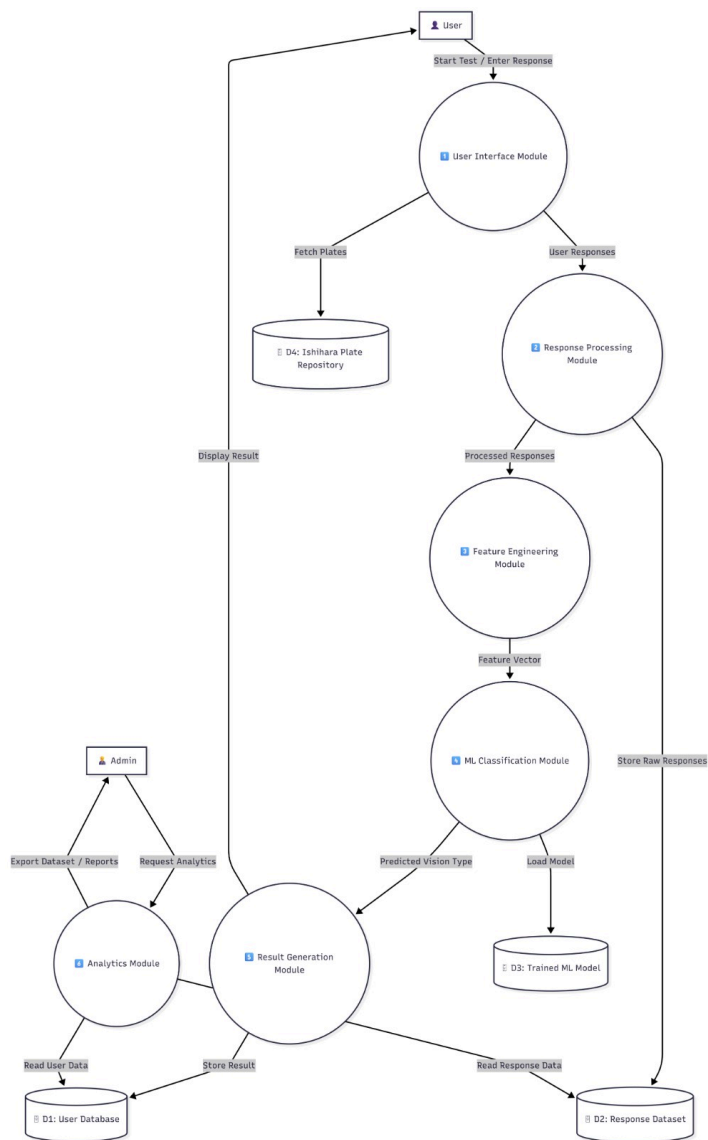
The sequence illustrates how user inputs are transmitted from the React-based frontend to the Flask backend, where responses are validated and prepared for prediction. The backend then forwards the processed data to the vision classification engine, which applies trained machine learning models to generate prediction results. These results are stored in the database and returned to the frontend for display.



B.4 Data Flow Diagram

This diagram illustrates how user response data moves through different layers of the VisionCheck AI system. It shows the flow starting from frontend validation of Ishihara test inputs, followed by backend processing, feature preparation, machine learning inference, and generation of classification results. The diagram highlights how raw user responses are transformed into meaningful vision classification outputs.

The data flow begins when a user submits responses through the frontend interface. The backend validates and preprocesses the data before forwarding it to the vision classification engine. The machine learning model analyzes the prepared input features and produces prediction results, which are then stored in the database and presented to the user through the interface.



B.4 Use Case Diagram

